

52 Pitch Flow Assessment System — Complete Implementation Package

Date: November 26, 2025

Version: 1.0

Status:  Production Ready



What This Package Includes

This is the **complete 52 Pitch Flow Assessment system** for the BARRELS app, ready for Abacus/dev team integration.

Core Components

1. **TypeScript Types** — Complete data structure specification
 2. **DeepAgent Report Generator** — AI skill for generating assessment reports
 3. **React UI Component** — Coach-facing assessment interface
 4. **API Route Stubs** — Backend endpoints for assessment management
 5. **PDF Template** — Complete 6-page report structure
 6. **Integration Guide** — Step-by-step implementation roadmap
-



Documentation Files

1. TypeScript Types

File: `types/assessment.ts`

Size: Medium (~5 pages)

Contents:

- Complete TypeScript interfaces for 52 Pitch Flow Assessment
- `AssessmentSession`, `PlayerInfo`, `DeviceFlags`, `SwingRecord`, `FlowScoresSummary`
- Helper functions: `getBandFromScore()`, `getBandLabel()`
- Type-safe data structures for all assessment components

Use this for: Type safety across assessment system

2. DeepAgent Report Generator

File: `docs/coach-rick-assessment-report-prompt.md`

Skill Name: `CoachRick.AssessmentReport`

Size: Large (~8 pages)

Contents:

- Complete DeepAgent system prompt for report generation
- 7-section output format (Snapshot, Card, Breakdown, Strengths, Bottlenecks, Plan, Re-Test)

- Input schema with all required fields
- Style guidelines and Coach Rick voice
- API call structure and usage examples
- Testing checklist for quality assurance

Use this for: Generating comprehensive assessment reports from 52-swing data

When to use:

- ☒ You have a completed 52-pitch assessment
 - ☒ You need comprehensive coaching roadmap
 - ☒ You're providing formal feedback
 - ☒ You want detailed Flow Path analysis
-

3. React UI Component

File: `app/assessments/52-pitch/run/page.tsx`

Size: Medium (~6 pages)

Contents:

- Complete coach-facing UI for running assessments
- Player selection, device toggles, progress tracking
- Real-time swing logging with progress bar
- Assessment completion and report generation
- BARRELS brand styling with gold theme
- Framer Motion animations for smooth UX

Use this for: Coach interface to run and monitor 52-pitch assessments

Features:

- Player dropdown (TODO: populate from real roster)
 - Device flags (motion, ball, sensor, neural)
 - Progress tracking (0/52 swings)
 - Manual swing logging button
 - Complete button with confirmation
 - Navigation to generated report
-

4. API Route Stubs ★

Files:

- `app/api/assessments/52-pitch/start/route.ts`
- `app/api/assessments/52-pitch/[sessionId]/swing/route.ts`
- `app/api/assessments/52-pitch/[sessionId]/complete/route.ts`

Size: Small (~3 pages total)

Contents:

- POST `/api/assessments/52-pitch/start` — Create new assessment session
- POST `/api/assessments/52-pitch/[sessionId]/swing` — Log a swing
- POST `/api/assessments/52-pitch/[sessionId]/complete` — Complete and generate report

TODO Items Marked:

- Database session creation logic
- Swing data storage/aggregation
- Summary score calculation
- Coach Rick AI report generation
- Report storage and retrieval

Use this for: Backend API integration points

5. PDF Report Template

File: docs/52-pitch-assessment-pdf-template.md

Size: Large (~12 pages)

Contents:

- Complete 6-page PDF structure
- Template variables for data binding
- Design guidelines (typography, colors, spacing)
- Layout examples for each page
- React-PDF and PDFKit implementation examples
- Testing checklist

Use this for: Generating printable/shareable assessment reports

Pages:

1. Cover & Snapshot (player info, big score)
 2. Momentum Transfer Card (5 score tiles)
 3. Flow Breakdown (Ground/Power/Barrel analysis)
 4. Strengths & Bottlenecks (2-4 strengths, 2-3 bottlenecks)
 5. Coaching Plan (theme, cues, drills, goals)
 6. Next Test & Notes (re-test timeline, handwritten notes area)
-

6. This Integration Guide

File: docs/52-pitch-assessment-complete.md

Size: Comprehensive (~15 pages)

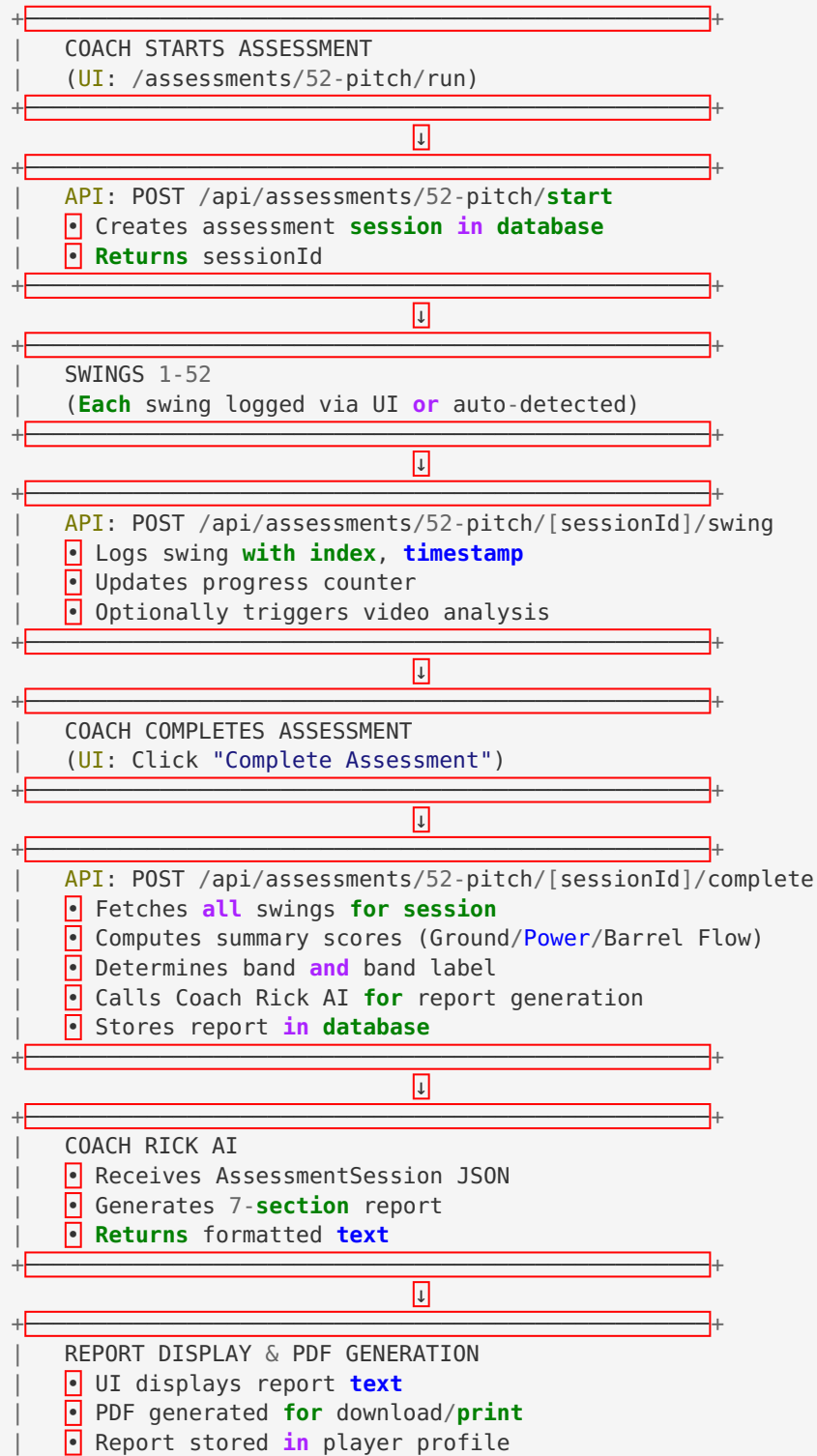
Contents:

- Complete package overview
- File manifest and descriptions
- Implementation roadmap
- Integration steps
- Testing checklist
- API flow diagrams
- Troubleshooting guide

Use this for: Master reference for 52 Pitch Assessment system

System Architecture

Data Flow



Implementation Roadmap

Phase 1: Database Setup (Week 1)

Goal: Prepare database to store assessment sessions and swing data

Tasks:

1. Update Prisma Schema

```

```prisma
model AssessmentSession {
 id String @id @default(cuid())
 userId String
 user User @relation(fields: [userId], references: [id])

 sessionType String // "assessment"
 assessmentKind String // "52_pitch_flow"
 version String @default("52_pitch_flow_1.0")

 swingsPlanned Int @default(52)
 swingsCompleted Int @default(0)
 completed Boolean @default(false)

 // Device flags
 useKinetrax Boolean @default(true)
 useHitTrax Boolean @default(true)
 useSensor Boolean @default(true)
 useNeuralTest Boolean @default(false)

 // Timestamps
 startedAt DateTime @default(now())
 endedAt DateTime?
 completedAt DateTime?

 // Summary scores (computed after completion)
 momentumTransferScore Int?
 groundFlowScore Int?
 powerFlowScore Int?
 barrelFlowScore Int?
 consistencyScore Int?
 movementQualityScore Int?
 neuralScore Int?
 ballContactScore Int?
 band Int? // -3 to +3
 bandLabel String?

 // Coach notes
 notesFromCoach String?
 contextTags String[] @default([])

 // Generated report
 reportText String?
 reportPdfUrl String?

```

```

// Relations
swings SwingRecord[]

createdAt DateTime @default(now())
updatedAt DateTime @updatedAt
}

model SwingRecord {
id String @id @default(cuid())
assessmentSessionId String
assessmentSession AssessmentSession @relation(fields: [assessmentSessionId], references: [id])

```

```

index Int // 1-52
timestamp DateTime @default(now())

// Pitch info
pitchType String?
pitchSpeed Float?
pitchLocationZone String?

// Motion metrics (JSON)
motionMetrics Json?

// Ball metrics (JSON)
ballMetrics Json?

// Sensor metrics (JSON)
sensorMetrics Json?

// Neural metrics (JSON)
neuralMetrics Json?

// Video reference
videoId String?
videoUrl String?

createdAt DateTime @default(now())
updatedAt DateTime @updatedAt
}
...

```

### 1. Run Migration

```

bash
cd /home/ubuntu/barrels_pwa/nextjs_space
yarn prisma migrate dev --name add_52_pitch_assessment
yarn prisma generate

```

---

## Phase 2: API Implementation (Week 1-2)

**Goal:** Implement backend logic for assessment management

**Tasks:**

1. **Implement** `/api/assessments/52-pitch/start`
    - Create `AssessmentSession` record
    - Store device flags
    - Return session ID
  2. **Implement** `/api/assessments/52-pitch/[sessionId]/swing`
    - Create `SwingRecord` entry
    - Update `swingsCompleted` counter
    - Optionally trigger video analysis
  3. **Implement** `/api/assessments/52-pitch/[sessionId]/complete`
    - Fetch all swings for session
    - Aggregate metrics (average Ground/Power/Barrel Flow)
    - Compute summary scores
    - Determine band and band label
    - Call Coach Rick AI
    - Store report
    - Return report ID
  4. **Create Report Retrieval Endpoint**

```
typescript
// GET /api/assessments/52-pitch/[sessionId]/report
// Returns the generated report text and PDF URL
```
- 

**Phase 3: Coach Rick AI Integration (Week 2)**

**Goal:** Configure DeepAgent skill and integrate with API

**Tasks:**

1. **Configure DeepAgent Skill**
  - Skill Name: `CoachRick.AssessmentReport`
  - Model: `gpt-4o`
  - Temperature: `0.7`
  - Max Tokens: `2000`
  - Paste system prompt from `docs/coach-rick-assessment-report-prompt.md`
2. **Create Report Generator Function**

```
``typescript
// lib/assessment-report-generator.ts
```

```
import type { AssessmentSession } from '@types/assessment';

export async function generateAssessmentReport(
 assessmentData: AssessmentSession
): Promise {
 const response = await fetch('https://apps.abacus.ai/v1/chat/completions', {
 method: 'POST',
 headers: {
 'Content-Type': 'application/json',
```

```

Authorization: Bearer ${process.env.ABACUSAI_API_KEY} ,
},
body: JSON.stringify({
 model: 'gpt-4o',
 messages: [
 {
 role: 'system',
 content: ASSESSMENT_REPORT_PROMPT,
 },
 {
 role: 'user',
 content: Generate assessment report:\n\n${JSON.stringify(assessmentData, null, 2)} ,
 },
],
 temperature: 0.7,
 max_tokens: 2000,
}),
});

```

```

const data = await response.json();
return data.choices[0].message.content;

```

```

}
```

```

1. Integrate with Complete Endpoint

- Call `generateAssessmentReport()` after computing summary
- Store returned report text in database

Phase 4: PDF Generation (Week 2-3)

Goal: Generate downloadable/printable PDF reports

Tasks:

1. Choose PDF Library

- Option A: `@react-pdf/renderer` (React components)
- Option B: `pdfkit` (Node.js library)
- Option C: Gamma API (external service)

2. Implement PDF Template

- Use structure from `docs/52-pitch-assessment-pdf-template.md`
- Create 6-page layout
- Apply BARRELS branding

3. Create PDF Generation Function

```

```typescript
// lib/pdf-generator.ts

```

```

export async function generateAssessmentPDF(
 assessmentData: AssessmentSession,

```



```

reportText: string
): Promise {
 // Generate PDF
 // Upload to S3
 // Return URL
}
```

```

1. Integrate with Complete Endpoint

- Call `generateAssessmentPDF()` after report generation
- Store PDF URL in database

Phase 5: UI Enhancement (Week 3)

Goal: Polish coach-facing UI and add report viewing

Tasks:

1. Populate Player Dropdown

- Replace mock player list with real roster
- Add search/filter functionality

2. Auto-Detect Swings (Optional)

- Integrate with video upload system
- Auto-increment swing counter when video is analyzed

3. Create Report View Page

```

typescript
// app/assessments/52-pitch/[sessionId]/report/page.tsx
// Displays the generated report text
// Provides PDF download button
// Shows summary scores in cards

```

4. Add Navigation

- Link from dashboard to assessment UI
- Link from player profile to past assessments

Phase 6: Testing & Refinement (Week 3-4)

Goal: Validate system end-to-end and refine based on feedback

Tasks:

1. Test Complete Flow

- [] Start assessment
- [] Log 52 swings (or fewer with confirmation)
- [] Complete assessment
- [] Verify report generation
- [] Verify PDF generation
- [] Verify report display

2. Test Edge Cases

- [] Partial assessment (< 52 swings)
- [] Missing data sources (no sensor, no neural)
- [] Different player levels (Youth, HS, College, Pro)
- [] Multiple assessments for same player

3. Refine AI Prompts

- Review generated reports for quality
- Adjust prompts if language is too technical
- Ensure Coach Rick voice is consistent

4. Performance Optimization

- Optimize summary score calculation
- Cache report generation if needed
- Optimize PDF generation time

Testing Checklist

Database

- [] AssessmentSession model created
- [] SwingRecord model created
- [] Relations properly defined
- [] Migrations run successfully
- [] Prisma client generated

API Endpoints

- [] POST /api/assessments/52-pitch/start works
- [] POST /api/assessments/52-pitch/[sessionId]/swing works
- [] POST /api/assessments/52-pitch/[sessionId]/complete works
- [] GET /api/assessments/52-pitch/[sessionId]/report works
- [] Authentication properly enforced
- [] Error handling implemented

Coach Rick AI

- [] DeepAgent skill configured
- [] Report generation function works
- [] 7-section format is consistent
- [] Coach Rick voice is maintained
- [] No mention of GOATY/S2/external brands
- [] Language is kid-friendly

PDF Generation

- [] PDF library integrated
- [] 6-page layout implemented
- [] BARRELS branding applied
- [] Score color coding works
- [] PDF uploads to S3 (or equivalent)

- [] PDF download works

UI

- [] Assessment start page loads
- [] Player dropdown populated
- [] Device toggles work
- [] Progress bar updates
- [] Manual swing logging works
- [] Complete button works
- [] Report page displays correctly
- [] PDF download button works
- [] Navigation works

End-to-End

- [] Complete 52-pitch assessment
- [] Report generated successfully
- [] PDF generated successfully
- [] Report displays correctly
- [] PDF is printable
- [] PDF is shareable



Troubleshooting Guide

Issue: Assessment session not creating

Symptoms:

- Error when clicking "Start Assessment"
- 500 error in network tab

Solutions:

1. Check database connection
 2. Verify Prisma schema is up to date
 3. Check API route logs for errors
 4. Verify user authentication
-

Issue: Swings not logging

Symptoms:

- Progress bar not updating
- Swing count stays at 0

Solutions:

1. Check sessionId is being passed correctly
 2. Verify API route is receiving requests
 3. Check database write permissions
 4. Verify swing data structure
-

Issue: Report not generating

Symptoms:

- Assessment completes but no report
- Report text is empty

Solutions:

1. Check Abacus.AI API key is set
 2. Verify Coach Rick skill is configured
 3. Check API rate limits
 4. Review API request/response in logs
 5. Verify assessment data structure matches expected format
-

Issue: PDF not generating

Symptoms:

- Report displays but PDF button doesn't work
- PDF URL is null

Solutions:

1. Check PDF library is installed
 2. Verify S3 credentials are set
 3. Check S3 upload permissions
 4. Review PDF generation logs for errors
 5. Verify template structure
-

Issue: Scores are incorrect

Symptoms:

- Summary scores don't match swing data
- Band label is wrong

Solutions:

1. Review score aggregation logic
 2. Check for null/undefined values in swing data
 3. Verify band mapping thresholds
 4. Review calculation formulas
 5. Test with known data set
-



Example Data

Sample Assessment Session

```
{
  "sessionId": "sess_52_2025_02_01_teen01",
  "sessionType": "assessment",
  "assessment": {
    "isAssessment": true,
    "assessmentKind": "52_pitch_flow",
    "swingsPlanned": 52,
    "swingsCompleted": 52,
    "completed": true
  },
  "version": "52_pitch_flow_1.0",
  "player": {
    "playerId": "player_teen01",
    "firstName": "Jalen",
    "lastName": "Brown",
    "age": 14,
    "level": "Youth",
    "handedness": "R",
    "bats": "R",
    "throws": "R"
  },
  "coach": {
    "coachId": "coach_rick",
    "name": "Coach Rick"
  },
  "devices": {
    "useKinetrax": true,
    "useHitTrax": true,
    "useSensor": true,
    "useNeuralTest": true
  },
  "startedAt": "2025-02-01T13:05:00Z",
  "endedAt": "2025-02-01T13:40:00Z",
  "summary": {
    "momentumTransferScore": 78,
    "groundFlowScore": 74,
    "powerFlowScore": 80,
    "barrelFlowScore": 79,
    "consistencyScore": 76,
    "movementQualityScore": 80,
    "neuralScore": 71,
    "ballContactScore": 77,
    "band": 1,
    "bandLabel": "Above Average"
  },
  "notesFromCoach": "First big assessment of the year. Good mover, needs a bit more ground stability and better late decisions.",
  "contextTags": [
    "first_assessment",
    "off_season"
  ]
}
```

Summary

What You Have

- ✓ **Complete TypeScript types** for 52 Pitch Flow Assessment
- ✓ **DeepAgent skill** for generating comprehensive reports
- ✓ **React UI component** for coach-facing assessment interface
- ✓ **API route stubs** with TODO markers for backend integration
- ✓ **PDF template** with 6-page structure and design guidelines
- ✓ **Implementation roadmap** with phased approach
- ✓ **Testing checklist** for quality assurance
- ✓ **Troubleshooting guide** for common issues
- ✓ **Example data** for testing and validation

What's Next

1. **Week 1:** Database setup + API implementation
2. **Week 2:** Coach Rick AI integration + PDF generation
3. **Week 3:** UI polish + report viewing
4. **Week 4:** Testing, refinement, and launch

File Manifest

| | |
|--|---|
| barrels_pwa/nextjs_space/ | |
|  types/ | |
|   assessment.ts |  TypeScript types |
|  docs/ | |
|   coach-rick-assessment-report-prompt.md |  DeepAgent prompt |
|   52-pitch-assessment-pdf-template.md |  PDF template |
|   52-pitch-assessment-complete.md |  This file |
|  app/ | |
|   assessments/52-pitch/run/page.tsx |  Coach UI |
|   api/assessments/52-pitch/ | |
|    start /route.ts |  Start endpoint |
|   [sessionId]/ | |
|   swing/route.ts |  Log swing endpoint |
|   complete/route.ts |  Complete endpoint |
|  lib/ | |
|   assessment-report-generator.ts |  AI report generator (TODO) |
|   pdf-generator.ts |  PDF generator (TODO) |

Status:  Ready for Development Team

Last Updated: November 26, 2025

Version: 1.0

 **Everything you need to integrate the 52 Pitch Flow Assessment system into BARRELS is ready!**